

Divya Kiran Kadiyala

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385 River Oaks Pkwy, Apt 2098, San Jose, CA 95134.

SUMMARY

I am a computer systems researcher focused on developing tailored system architectures for parallel, scientific, and deep learning AI/ML workloads in High Performance Computing environments. My research spans computer architecture, memory system design, distributed AI/ML systems, and CXL-based technologies, with focus on accelerating memory bandwidth-intensive applications in resource-constrained environments. I am always open to opportunities and collaborations where I can apply my academic expertise and industry experience to drive innovation in next-generation memory and computer system architectures.

EDUCATION

Georgia Institute of Technology (Georgia Tech), Atlanta, GA 08/19/2019 – 12/11/2025
Doctor of Philosophy (PhD) in *Electrical and Computer Engineering*

Thesis: Memory system optimizations for parallel and bandwidth-intensive applications

Advisor: [Prof. Alexandros Daglis](#)

Arizona State University (ASU), Tempe, AZ 08/11/2015 – 05/08/2017
Master of Science in Engineering (MSE) in *Electrical Engineering*
Specialization: VLSI and Mixed Signal Circuits

K L University (KLU), Guntur, A.P., India 07/01/2009 – 04/30/2013
Bachelor of Technology (B.Tech) in *Electronics and Communication Engineering*

PROFESSIONAL EXPERIENCE

Hewlett Packard Enterprise (HPE), 820 N. McCarthy Blvd, Milpitas, CA 95035

- **AI Performance Architect**

HPC Advanced Dev Organization, HPE 02/09/2026 – Present

- Building high-fidelity **performance models** and **emulation platforms** to validate SoC subsystems, memory hierarchies, and multi-node architectures, guiding system-level design decisions for improving the performance of HPC-AI workloads on large-scale distributed systems.
- Leading **collaborative R&D efforts** with national laboratories, academic institutions, and industry partners to evaluate architectural trade-offs across compute, memory, and interconnect subsystems for next-generation HPC & AI supercomputing systems.
- Designed composable AI system architectures using **CXL-based shared-memory fabrics** and **disaggregated memory expansion** to enable scalable memory expansion, efficient distributed data exchange, and improved hardware utilization for AI training and inference.
- Driving **intellectual property** creation by identifying patentable innovations in hardware acceleration and distributed-system methodologies and supporting invention disclosures/filings.
- Submitted research papers to top-tier **systems and architecture conferences** presenting advances in computer architecture, memory systems, and high-performance computing.

- **Research Associate Intern - Remote** (Part-term CPT)

Network and Distributed Systems Lab (NDSL), HPE Labs

- Phase I: 08/26/2024 – 05/01/2025
- Phase II: 05/12/2025 – 07/31/2025
- Phase III: 08/25/2025 – 12/11/2025
- Engineered a novel distributed training architecture for AI/ML workloads utilizing **disaggregated memory expansion** to decouple compute and memory resources, significantly improving hardware utilization.

- Designed and implemented a **CXL-based (Compute Express Link)** shared memory fabric to facilitate seamless memory expansion and efficient data exchange across distributed nodes, optimizing latency using software-hardware co-design.
- Conducted rigorous performance characterization of memory-bound AI workloads, identifying bottlenecks in traditional PCIe-based systems and proposing CXL-driven solutions.
- **HPE TechCon:** Selected to present research on "CXL Shared Memory System for Distributed AI/ML Training" at HPE's premier internal technical conference; paper was peer-reviewed and accepted by a committee of HPE Distinguished Technologists.
- Co-invented a proprietary architectural and system framework for distributed AI training using CXL based shared memory; Patent filed through HPE Intellectual Property department.

• **Research Associate Intern (Full-term CPT)**

Network and Distributed Systems Lab (NDSL), HPE Labs

05/13/2024 – 08/16/2024

- Architected a novel **Hyper-Collective Framework** for infrastructure sharing, enabling high-utilization execution of AI/ML workloads across multi-tenant supercomputing fabrics.
- Modeled and mitigated network communication bottlenecks by optimizing **collective communication algorithms** (e.g., All-Reduce, All-to-All) for complex multi-tenant system topologies.
- Engineered low-level performance enhancements using **CUDA and ROCm** kernel optimizations, implementing an **algorithm-topology co-design** to maximize collective communication throughput in heterogeneous GPU environments.
- Presented **Hyper-Collective Framework** poster at **HPE Annual Intern Project Fair**; research reviewed by senior technical staff for impact on HPC infrastructure sharing

Samsung Semiconductor Inc. (SSI), 3655 North First Street, San Jose, CA 95134

• **Systems Technology Research Intern (Full-term CPT)**

Memory Solutions Lab (MSL)

05/09/2022 – 08/19/2022

- Engineered **CXL-enabled Memory Pooling** solutions to optimize resource utilization for high-performance computing (HPC) and cloud infrastructure
- Architected and evaluated novel **memory fabric topologies** based on the **CXL 3.0 specification**, focusing on distributed system architectures.
- Built analytical **performance models** to quantify the impact of memory expansion on large-scale AI/ML and distributed cloud workloads.

Luminous Computing, 278 Castro St, Mountain View, CA 94041

• **CPU Architecture Intern (Full-term CPT)**

05/24/2021 – 08/13/2021

- Built cycle-accurate **performance models** for **RISC-V microarchitectures** to analyze instruction execution, pipeline behavior, and throughput/latency trade-offs.
- Performed architectural evaluation and design-space exploration for high-performance **RISC-V-based AI inference accelerators**, guiding microarchitectural and system-level optimizations.
- Engineered **SystemC Transaction-Level Models (TLM)** to simulate and validate proposed microarchitecture enhancements for next-generation RISC-V processors.

Cadence Design Systems, 2655 Seely Avenue, San Jose, CA 95134

• **Sr. Applications Engineer (Full-term CPT)**

07/10/2017 – 07/31/2019

- Developed and deployed advanced **Electromigration (EM)** and **IR-drop** solution flows for static and dynamic power integrity analysis using the **Cadence Voltus** toolset.
- Led end-to-end **power sign-off** execution for complex SoC tapeouts, ensuring reliability and performance across multiple process nodes.
- Owned the design and rollout of **Power-Grid View (PGV)** generation flows to enable efficient hierarchical power analysis for a major Bay Area silicon hardware designer.
- Partnered with R&D to define requirements and to architect, validate, and productize new capabilities in **EDA** tools, translating customer needs into robust tool functionality.
- Authored technical documentation and delivered targeted training to help engineering teams adopt advanced power-aware design methodologies.

Tata Consultancy Services, 185 Lloyds Road, Chennai, Tamil Nadu, India 600086

- **Assistant Systems Engineer (Full-time)** 10/23/2013 – 04/30/2015
 - Administered and troubleshoot **SAP Roles and Authorizations**, providing timely access support and resolving security-related incidents for enterprise users.
 - Conducted periodic **access reviews** and supported **audit** activities to ensure compliance with organizational information security policies and controls.
 - Delivered end-user support for **user provisioning, deprovisioning, and access changes**, maintaining accurate documentation and adhering to established SLAs.
 - Prepared and presented **process documentation** and reports for business stakeholders to review, improving clarity and consistency of access-governance workflows.

RESEARCH AND TEACHING EXPERIENCE

Georgia Institute of Technology, 225 North Ave. NW, Atlanta, GA 30332

- **Graduate Research Assistant (GRA)** 05/01/2020 – 12/11/2025
Networked Architectures and Systems Lab (NArSyL), School of Computer Science (SCS), Georgia Tech
Advisor: [Prof. Alexandros Daglis](#)

Conducted doctoral research in computer architecture and systems with emphasis on **memory-system optimizations** for parallel and bandwidth-intensive workloads, **AI/ML performance modeling**, **distributed training**, **high-performance computing (HPC)**, and **CXL-based memory technologies**. Developed simulation-driven evaluation methodologies and hardware–software techniques to improve performance and efficiency for modern datacenter, cloud, and HPC workloads. Published peer-reviewed work in top-tier computer architecture and systems venues.

Thesis: Memory system optimizations for parallel and bandwidth-intensive applications

1. Acceleration of HPC workloads on Hardware Transactional Memory (HTM) [*HPCA'23*]
 - **Problem:** Capacity aborts leads to severe performance loss reducing the utility of commercial HTMs.
 - **Insight:** Track only the critical memory accesses to increase the effective capacity of on-chip HTM buffers.
 - **Contribution:** Developed **HinTM**, a novel hardware–software co-design technique that leverages software hints to track only critical memory accesses within a transaction.
 - **Result:** Achieved up to 8.7× speedup over baseline HTM by eliminating 64% of transactional capacity aborts.
2. Acceleration of memory-bound server workloads using I/O bandwidth harvesting [*under peer review*]
 - **Problem:** Reduced per-core bandwidth in server CPUs leads to high memory latency and queuing delays.
 - **Insight:** Augment the memory bandwidth by dynamically harvesting idle I/O bandwidth in many-core CPUs.
 - **Contribution:** Developed **SURGE**, a software-assisted architectural mechanism that opportunistically harvests unused I/O bandwidth to access additional memory via high-speed serial links, such as CXL.
 - **Result:** Reduced memory queuing delay by 33% and achieved up to 1.5× speedup over a DDR-only baseline.
3. Acceleration of distributed AI training workloads on large-scale clusters [*arXiv*]
 - **Problem:** Increased model sizes force cluster scale-out and cause slowdowns from communication bottlenecks.
 - **Insight:** Expanding memory capacity in training nodes limits scale-out requirements improving throughput.
 - **Contribution:** Developed **COMET**, a holistic cluster design methodology for rapid design-space co-exploration to evaluate the impact of memory expansion on distributed deep learning training performance.
 - **Result:** Identified viable designs for memory expansion to improve training throughput for LLMs and DLRMs.

CPT Note: During **Summer 2021**, **Summer 2022**, and **Summer 2024**, I completed full-time internships authorized under CPT; during these periods, I was not employed as a Graduate Research Assistant at Georgia Tech.

- **Graduate Teaching Assistant (GTA)**

Served as Graduate Teaching Assistant for the following courses at Georgia Tech.

- **High Performance Computer Architecture** (CS 6290 / ECE 6100) Summer 2023, Fall 2021
- **VLSI and Advanced Digital Design** (ECE 3150), Georgia Tech Spring 2020
- **Digital System Design** (ECE 2020), Georgia Tech Fall 2019

- Completed **CETL 8000** GTA preparatory training at Georgia Tech.
- Supported course delivery by preparing instructional materials, conducting **recitation sessions** and **office hours**, and supervising **laboratory sessions**.
- Graded assignments, quizzes, and exams, providing timely feedback and maintaining consistent evaluation standards.
- Designed and maintained digital-logic laboratory and exercises using **Logisim-generic-2.7.1**, and assisted students with debugging and design verification.

Arizona State University, 1151 S. Forest Ave., Tempe, AZ 85287

- **Volunteer Research Aide** 07/01/2016 – 07/07/2017
VLSI Research Lab, School of Electrical, Computer and Energy Engineering (ECEE), ASU
- Advisor:** Dr. Lawrence T. Clark, Emertius Professor, School of ECEE, ASU
 - Worked on novel SRAM based Hardware **Physically Unclonable Functions** (PUFs) and **True Random Number Generators**.
 - Published two research papers in top-tier VLSI and Circuits journals (TVLSI (2018) and TCAS (2019)).

SELECTED PUBLICATIONS

1. **Harvesting idle I/O resources for boosting memory bandwidth** [*under peer review*]
D. K. Kadiyala, and A. Daglis
2. **Enabling Flexible and Composable AI Systems via Memory Disaggregation** [*under peer review*]
D. K. Kadiyala, L. Cao, P. Sharma, S. Sury, and A. Daglis
3. **Geode: A Zero-shot Geospatial Question-Answering Agent with Explicit Reasoning and Precise Spatio-Temporal Retrieval**
D. Gupta, A. Ishaqui, and D. K. Kadiyala
ISCA Workshop Emerging Vision and Graphics System and Architectures (EVGA), June 2024
4. **COMET: A Comprehensive Cluster Design Methodology for Distributed Deep Learning Training**
D. K. Kadiyala, S. Rashidi, T. Heo, A. R. Bambhaniya, T. Krishna, and A. Daglis
preprint arXiv, 2022
5. **Safety Hints for HTM Capacity Abort Mitigation**
A. Jain*, D. K. Kadiyala*, and A. Daglis
High-Performance Computer Architecture (HPCA), 2023. Acceptance rate: 25.0%
* Equal Contribution
6. **Exploring Memory Expansion Designs for Training Mixture-of-Experts Models**
T. Heo, S. Rashidi, C. Man, D. K. Kadiyala, W. Won, S. Srinivasan, M. Elavazhagan, M. Kumar, A. Daglis, and T. Krishna
Workshop on Hot Topics in System Infrastructure, (HotInfra), June 2023
7. **Physically Unclonable Functions Using Foundry SRAM Cells**
L. T. Clark, S. B. Medapuram, D. K. Kadiyala, and J. Brunhaver
IEEE Transactions on Circuits and Systems I (TCAS), 2019. Acceptance rate: 30.0%
8. **SRAM Circuits for True Random Number Generation Using Intrinsic Bit Instability**
L. T. Clark, S. B. Medapuram, and D. K. Kadiyala
IEEE Transactions on Very Large Scale Integration Systems, (TVLSI), 2018. Acceptance rate: 37.3%

TECHNICAL SKILLS

Programming Languages : C, C++, CUDA, Perl, Python, System Verilog, Bash Scripting
Performance modeling : ZSim, ASTRA-Sim, DRAMSim, gem5, SESC, Garnet2.0
HPC & AI Tools : MPI, OpenMP, NCCL, Tensorflow, PyTorch, Keras
Processor Architectures : X86, RISC V, GPU, TPU, dataflow architectures
Logic Design : Verilog HDL, System Verilog, Cadence & Synopsys EDA tools

SPOKEN LANGUAGES

Telugu (native), Hindi (fluent), and English (fluent)